

Lab Exercise #10: Sorting A list Using Function Pointers

1 Overview

For this lab create and implement functions that use pointer functions. The file `Mathematicians.txt` contains a fictitious college class of mathematicians. Each record contains a first name, last name, yesar and age.

- Create a structure `struct student class[]` to hold the student information.
- Below is an example of how to read in a file. You'll need to make adjustments based on what you are reading in.

```
FILE *fptr;
```

```
// Open a file in read mode
```

```
fptr = fopen("filename.txt", "r");
```

```
// Store the content of the file
```

```
char myString[100];
```

```
// Read the content and store it inside myString
```

```
fgets(myString, 100, fptr);
```

```
// Print the file content
```

```
printf("%s", myString);
```

```
// Close the file
```

```
fclose(fptr);
```

- You will need to use the `qsort()` function provided in the `stdlib`. This function takes a pointer to the start of an array, the number of elements in the array and the size of each element. Additionally it takes a function pointer that has two parameters. A sample of using the `qsort()` function is provided below.

```
void qsort(void *arr, int num, int size, int (*fp)(void *pa, void *pb));
```

- The pointer function compares two elements within the array. Similar to `strcmp()`, this function is

required to return a negative, zero or positive depending on whether the first element (**pa**) are "less" than, equal to or "greater" than the element pointed to by the second pointer (**pb**).

- Write the appropriate `callback()` function to sort the records based on last name.
- Write the appropriate function that sorts and records based on first name.
- Write a function `void apply()` that iterates through the elements of the array calling a function for each element of the array.
- Write a function `isolder()` that prints the record if the age of the student is over 20 and does nothing otherwise.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from `header.c`). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #10' - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #10 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below. I COULD NOT FIT THE LINE WITH MY NAME AND THE LAB # ON THS SCREENSHOT. YOU SHOULD STILL HAVE THAT PRINT.

```
Raw records:
-----
Henri      Jacobi      2 21
Marie      Curie       4 25
Katherine  Johnson    3 22
Paul       Erdos       4 29
Carl       Gauss       3 32
Srinivasa  Ramanujan  3 17
Maria      Agnesi      4 25
Marjorie   Brown       1 17
Isaac      Newton      2 19
Leonard   Euler       2 19

Sorted by first name:
-----
Carl       Gauss       3 32
Henri      Jacobi      2 21
Isaac      Newton      2 19
Katherine  Johnson    3 22
Leonard   Euler       2 19
Maria      Agnesi      4 25
Marie      Curie       4 25
Marjorie   Brown       1 17
Paul       Erdos       4 29
Srinivasa  Ramanujan  3 17

Sorted by last name:
-----
Maria      Agnesi      4 25
Marjorie   Brown       1 17
Marie      Curie       4 25
Paul       Erdos       4 29
Leonard   Euler       2 19
Carl       Gauss       3 32
Henri      Jacobi      2 21
Katherine  Johnson    3 22
Isaac      Newton      2 19
Srinivasa  Ramanujan  3 17

People older than 20:
-----
Maria      Agnesi      4      25
Marie      Curie       4      25
Paul       Erdos       4      29
Carl       Gauss       3      32
Henri      Jacobi      2      21
Katherine  Johnson    3      22

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Lab#10_FirstName_LastName.c`

Due date: The lab is due at the end of your lab period, either Tuesday, November 11th or Thursday, November 13th.

Be sure to save a copy of your .c file.